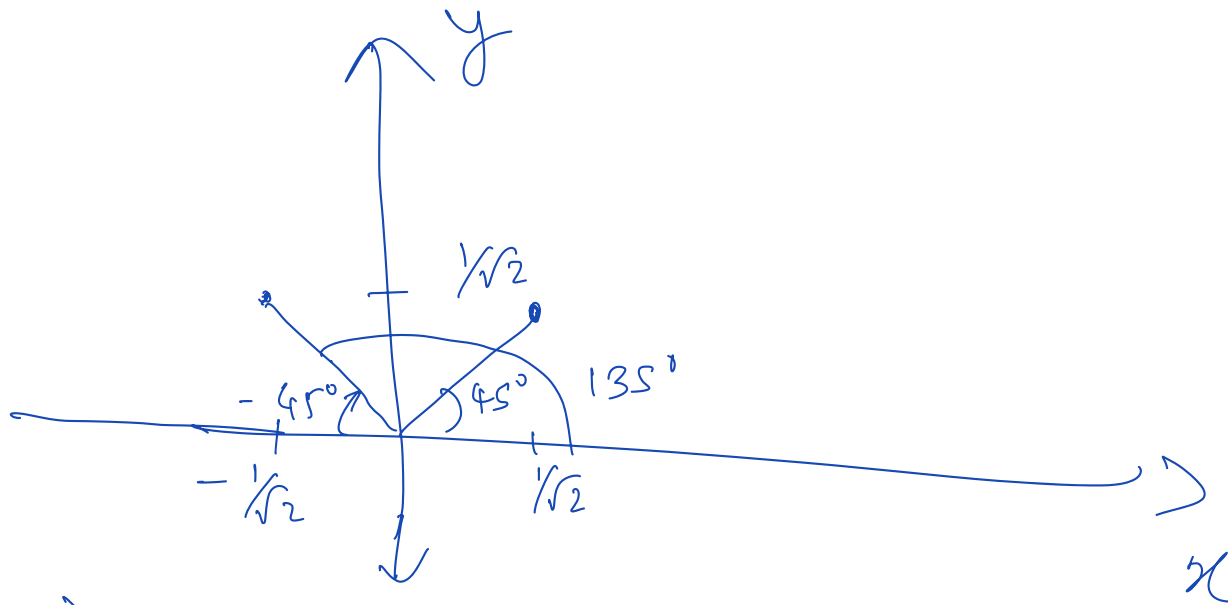


Lmt Class (Oct 20)

- 1) Motivation for high-dim viz
- 2) PCA introduction/intuition
- 3) Applying PCA
- 4) Geometrical Obj
- 5) Mathematical proof
- 6) Eigen values and vectors

Today's class

- 1) PCA Recap
- 2) Information loss in PCA
- 3) Code
- 4) High-dim viz using PCA
- 5) t-SNE
- 6) Problems with t-SNE



$$\text{eig vec } 1 = \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$$

$$\text{eig vec } 2 = -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$$

$$\begin{aligned} \tan^{-1} \left(\frac{1/\sqrt{2}}{1/\sqrt{2}} \right) \\ = \tan^{-1}(1) \\ = 45^\circ \end{aligned}$$

$$\tan 45^\circ = 1$$

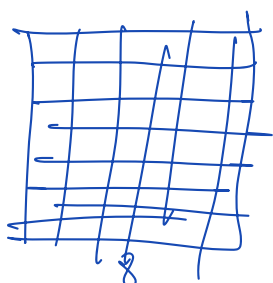
$1 \times 2 \rightarrow$ eigen vec $\rightarrow$ as row

$N \times 2 \rightarrow$ data

$$\text{data.T} \Rightarrow 2 \times n$$

projections of
orig data on
PC 1

$$\begin{array}{c} (1 \times 2) \\ \uparrow \\ \text{row eig vec} \end{array} \times \begin{array}{c} (2 \times n) \\ \downarrow \\ \text{data transpose} \end{array} \rightarrow \boxed{1 \times n} \rightarrow \begin{array}{c} \text{single} \\ \text{row} \end{array}$$



8

8x8 matrix

$$\boxed{V u = \lambda u}$$

$$(V - \lambda I) u = 0$$

↓

$$\lambda = 3, 3, -4$$

$$V - \lambda I = 0$$

↓

system of linear equations

↓

$$\lambda = 3, 3, -4$$

$$(\lambda - 3)^2 (\lambda + 4) = 0$$

1) Can we have duplicate eigen values?

2) if yes, will corresponding eigenvectors be repeated?

3) if yes, is there multi-collinearity problem?

UMAP → pre-tend

If $\lambda = 0$, does it mean multi-collinearity?
↳ m other PCs have already explained 100% of the data.